

Swaham Pitambar Patra

Program: Basic OOP Program

```
Tasks 30-06 > OOP > classCreation.py > ...  
1  class Student:  
2      name = 'Swaham'  
3  
4  S = Student()  
5  print(S.name)
```

Output:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python\DS\Tasks 30-06\OOP\classCreation.py"  
Swaham  
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS>
```

Program: __init__() Method

```
Tasks 30-06 > OOP > constructorCreation.py > ...
1  class Student:
2      branch = 'CO'
3      def __init__(self, name):
4          self.name = name
5
6  S = Student("Swaham")
7  print(S.name)
8  print(S.branch)
```

Output:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python\DS\Tasks 30-06\OOP\constructorCreation.py"
Swaham
CO
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS> |
```

Program: Classes and Objects

```
Tasks 30-06 > OOP > initUse.py > ...
1  class Person:
2      def __init__(self,name,age):
3          self.name = name
4          self.age = age
5      def showAge(self):
6          print(f"Hello! I'm a Person named {self.name},{self.age} years old.")
7
8  Person1 = Person("Swaham",18)
9  Person1.showAge()
10
11 Person2 = Person("Priya",23)
12 Person2.showAge()
```

Output:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 30-06\OOP\initUse.py"
Hello! I'm a Person named Swaham,18 years old.
Hello! I'm a Person named Priya,23 years old.
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> |
```

Program: Instance Variables

Tasks 30-06 > OOP > class_variable_use.py > ...

```
1  class Car:
2      wheels = 4 # class variable
3      def __init__(self,color):
4          self.color = color # instance variable
5
6  car1 = Car("red")
7  car2 = Car("blue")
8
9  print(f"{car1.color} and {car2.color}")
10 print(f"{Car.wheels}")
```

Output:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python\DS\Tasks 30-06\OOP\class_variable_use.py"
red and blue
4
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS>
```

Program: Instance vs Class Variables

```
Tasks 30-06 > OOP > C_vs_I_variable.py > Book > __init__
1  class Book:
2      shelves = 5  # class variable
3      def __init__(self, title):
4          self.title = title  # instance variable
5
6  Book1 = Book("Python 101")
7  Book2 = Book("Java Basics")
8
9  print(f"Books {Book1.title} and {Book2.title}")
10 print(f"Shelves of books {Book.shelves}")
```

Output:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python\OS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python\OS\Tasks_30-06\OOP\C_vs_I_variable.py"
Books: Python 101 and Java Basics
Shelves of books 5
PS C:\Users\Swaham Patra\OneDrive\ITR Python\OS>
```